

Central Asian University

INTERNSHIP DIARY

Student Name:	Ismoil Rahimov
Student ID:	230514
Programme:	Computer Science / IT
Company:	Elite Support Solutions
Department:	IT & Software Development
Duration:	01 March 2026 – 30 April 2026 (8 weeks)

Week 1 (01–07 March 2026)

Focus: Getting started and setting up

Mon 01 Mar	First day at Elite Support Solutions. My supervisor showed me around the office and introduced me to the IT team. Got my laptop and login details. Spent most of the day reading about how the company works — they manage a big fleet of trucks and a lot of drivers.
Tue 02 Mar	Set up Python 3.10, VS Code and OpenCV on the laptop. Also installed Git and made sure everything was working. Read through the company's rules about handling data.
Wed 03 Mar	My supervisor explained that the IT team sometimes needs to scan documents like driver licenses and vehicle stickers. That gave me the idea to focus my project on QR and barcode detection. Spent the afternoon reading OpenCV documentation.
Thu 04 Mar	Created the GitHub repository and set up the folder structure. Wrote a basic README. Nothing fancy yet but good to have it set up from the beginning.
Fri 05 Mar	Team meeting in the morning. We talked about what exactly the project should do. I wrote a short summary of the week for my supervisor.

Week 2 (08–14 March 2026)

Focus: Collecting data and reading papers

Mon 08 Mar	Started collecting test images. Took photos of QR codes I found on packages and printed some myself. Also photographed some barcodes on products in the office supply room.
Tue 09 Mar	Generated extra QR codes online to make sure I had enough variety — codes with URLs, phone numbers and plain text. Saved them all to the dataset folder.
Wed 10 Mar	Read a few articles about how QR code detection works. Took notes for the literature review. The main thing I learned is that the three corner squares are what makes detection possible.
Thu 11 Mar	Sorted my 50 images into three groups: easy clean ones, rotated ones, and difficult real-world photos. Labeled each one with what the expected output should be.
Fri 14 Mar	Showed the dataset to my supervisor. He suggested adding more images with bad lighting, so I went and took a few more photos in different conditions.

Week 3 (15–21 March 2026)

Focus: Writing the detection code

Mon 15 Mar	Started actually writing the detector. Used <code>cv2.QRCodeDetector</code> and the <code>detectAndDecodeMulti</code> function. Tested it on 5 images and it worked right away which was a nice surprise.
Tue 16 Mar	Wrote the function that draws the green box around detected codes and shows the text on screen. Had to figure out how the point arrays work to draw the polygon correctly.
Wed 17 Mar	Added the <code>process_image</code> function so the user can just give a file path and the program does everything. Results get saved automatically to a results folder.
Thu 18 Mar	Found a bug where empty detections were causing the program to crash. Fixed it by adding a simple if data: check before processing each result.
Fri 21 Mar	Cleaned up the code and added comments. Committed the clean version to GitHub. My supervisor looked at it and said it was good.

Week 4 (22–28 March 2026)

Focus: Webcam mode and testing

Mon 22 Mar	Added the webcam mode using <code>cv2.VideoCapture</code> . Also added a small FPS counter in the corner so you can see how fast it is running.
Tue 23 Mar	Tested the webcam mode with QR codes on my phone screen. Tried different distances and angles. It detected most of them without any trouble.
Wed 24 Mar	Measured performance. It runs at around 27 FPS on the office laptop which is more than enough for smooth video. The main slowdown is actually just the display function.
Thu 25 Mar	Wrote the batch evaluation function that runs the detector on a whole folder of images and prints a summary table. Ran it on all 50 test images.
Fri 28 Mar	Wrote an interim report for my university supervisor and uploaded all the code to GitHub.

Week 5 (29 Mar – 04 Apr 2026)

Focus: Analyzing results

Mon 29 Mar	Went through the results carefully. 47 out of 50 images were detected correctly. The 3 failures were all edge cases — one very blurry, one tiny code, one partially covered.
Tue 30 Mar	Calculated the metrics: 94% accuracy, 0% false positives, 27 FPS, about 36ms per image. Wrote them down for the report.

Wed 01 Apr	Made some annotated example images showing the green boxes and decoded text. These will go into the report as figures.
Thu 02 Apr	Discussed results with supervisor. He mentioned that a deep learning approach would handle the hard cases better. I added that as a future improvement idea.
Fri 04 Apr	Updated the GitHub README with the results table and clearer usage instructions.

Week 6 (05–11 April 2026)

Focus: Writing the report

Mon 05 Apr	Started writing the technical report. Did the abstract, introduction and literature review first since those are the most straightforward.
Tue 06 Apr	Wrote the problem statement, objectives and dataset sections. Added the tables for dataset breakdown and results.
Wed 07 Apr	Finished the methodology section with descriptions of each step in the pipeline. Included the main code snippet.
Thu 08 Apr	Completed results, metrics, conclusion and references. Read through the whole thing and fixed some awkward sentences.
Fri 11 Apr	Sent the draft to my supervisor for review. He had a few small corrections mainly about formatting.

Week 7 (12–18 April 2026)

Focus: Preparing the presentation

Mon 12 Apr	Applied the corrections from my supervisor. Final report compiled and saved as PDF.
Tue 13 Apr	Made the presentation slides. Tried to keep it to around 10-12 slides covering all the main points without going into too much detail.
Wed 14 Apr	Practiced the presentation with a timer. First run was about 16 minutes so I cut some slides down to get it closer to 12.
Thu 15 Apr	Did a practice run in front of two colleagues from the IT team. They asked a few questions which helped me prepare for the viva.
Fri 18 Apr	Made some small improvements to the slides based on the feedback. Uploaded everything to GitHub.

Week 8 (19–30 April 2026)

Focus: Final week and submission

Mon 19 Apr	Final check of all the code. Made sure every function has a proper docstring and the comments make sense.
Tue 20 Apr	Checked the GitHub repository structure one more time. All required folders are there with the right files.
Thu 24 Apr	Wrote the internship diary and the internship report. Took most of the day.
Fri 25 Apr	Exit meeting with my supervisor. He gave me a positive evaluation which I was happy about.
Wed 30 Apr	Submitted everything to the university. Internship finished.

Supervisor Confirmation & Company Stamp

Supervisor Signature
& Company Stamp:

